



Exploratory Data Analysis of Spotify Tracks Dataset

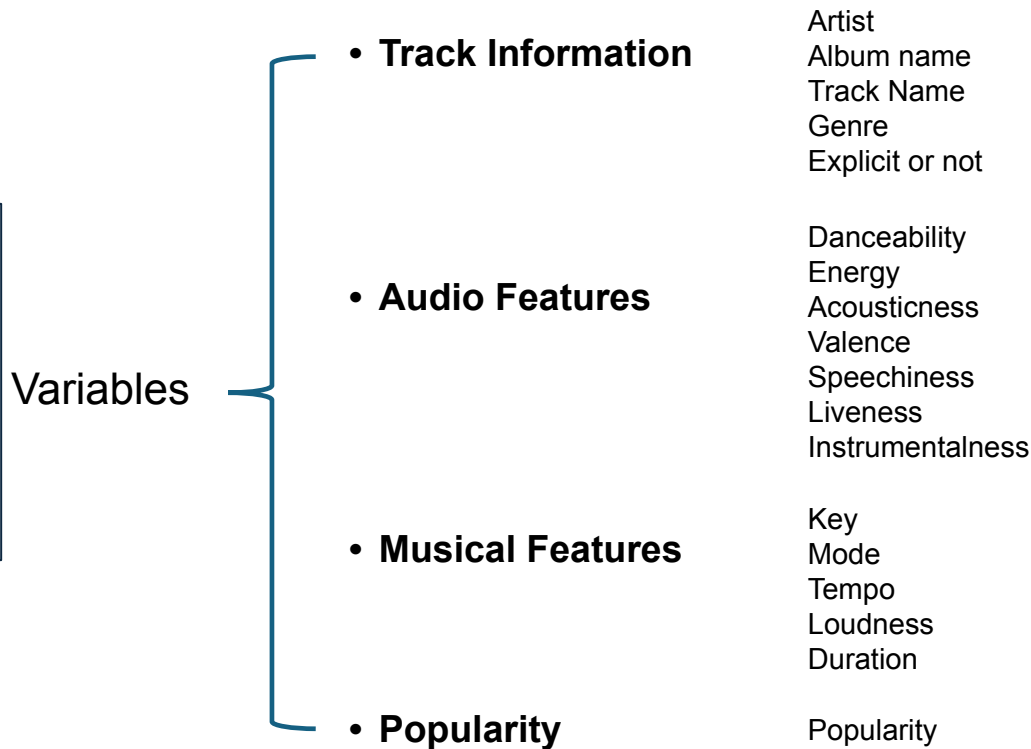
Team: Jianing Bao, Victor Grapin, Yoan Liu, Jenifar Mazumder

Date: 2026 - 07 - 03

Content

1. Dataset selection
2. Data cleaning
3. Univariate EDA
4. Bivariate EDA
5. Conclusion

Data Selection- Spotify Tracks (Kaggle)



Rows: 114000

Columns: 21

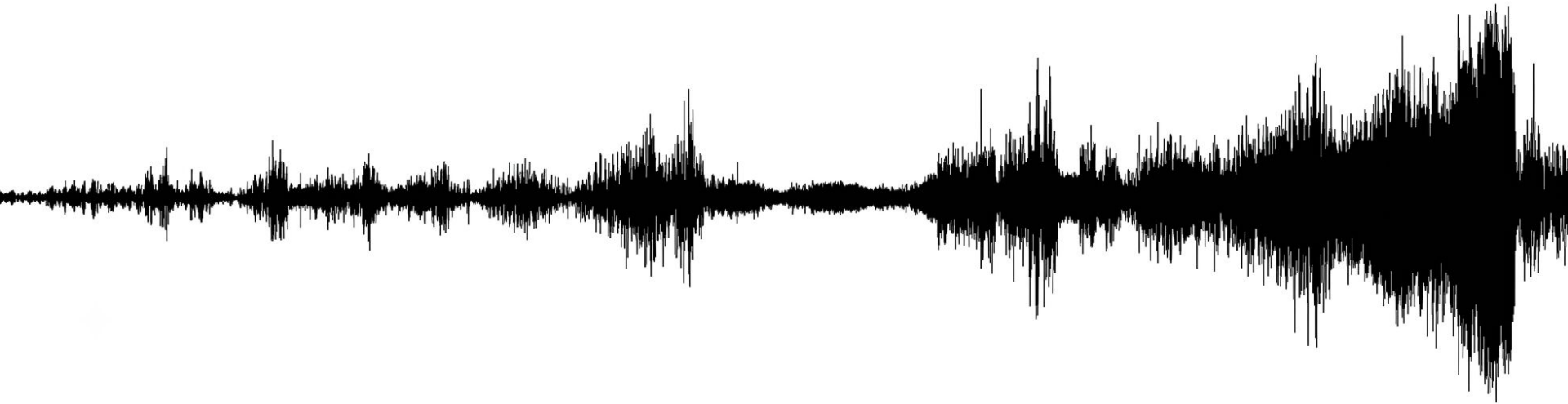
- ❖ Index columns: 1
- ❖ Categorical columns : 6
- ❖ Numerical columns:14

Data Cleaning

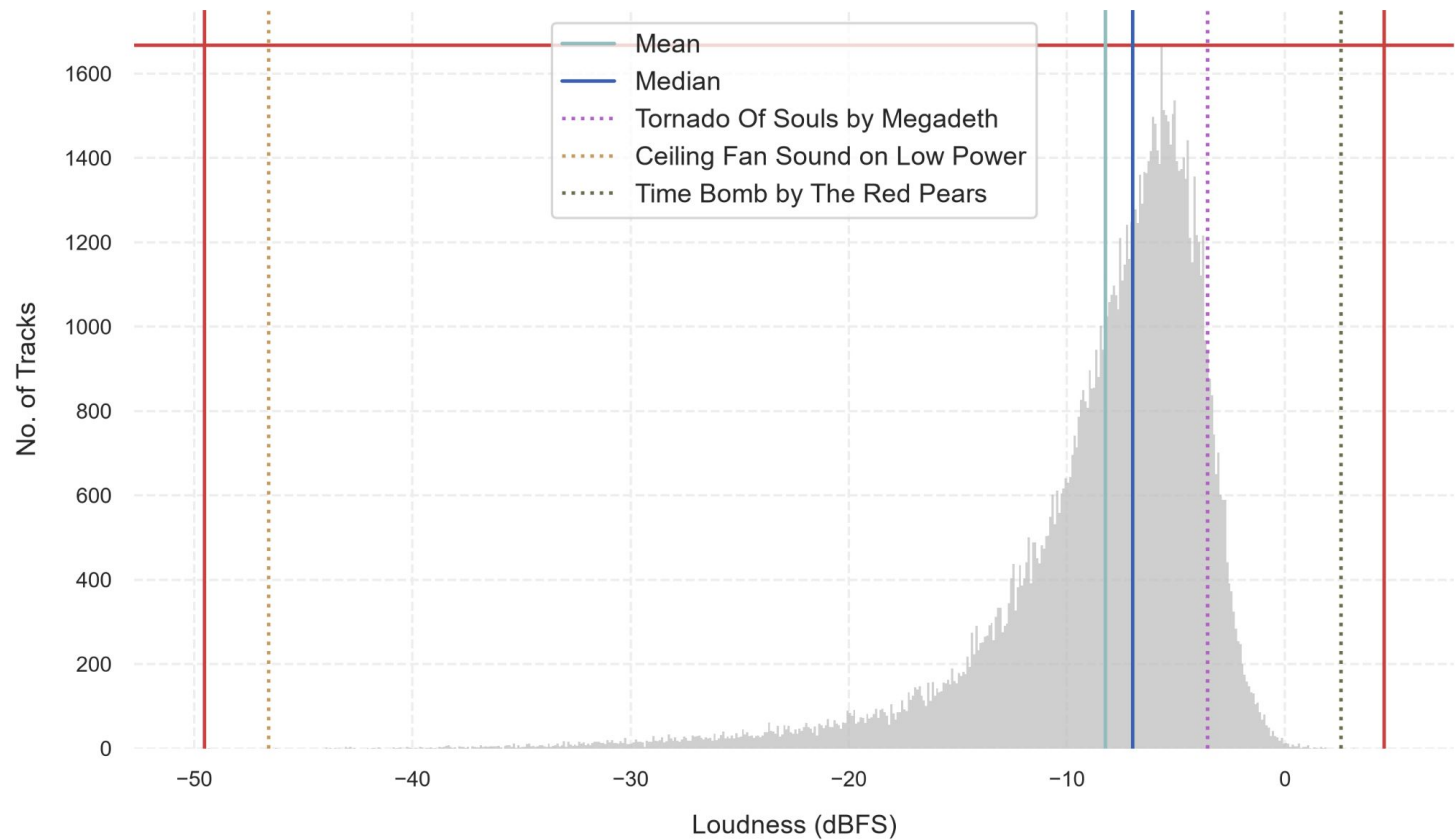
Steps:

1. Check the shape of the dataframe.
2. Assigned the Index column of the dataset to that of the DF.
3. Check for duplicate rows in the DF (450 duplicates found).
4. Drop the duplicates and missing values.
5. Save the cleaned dataset.

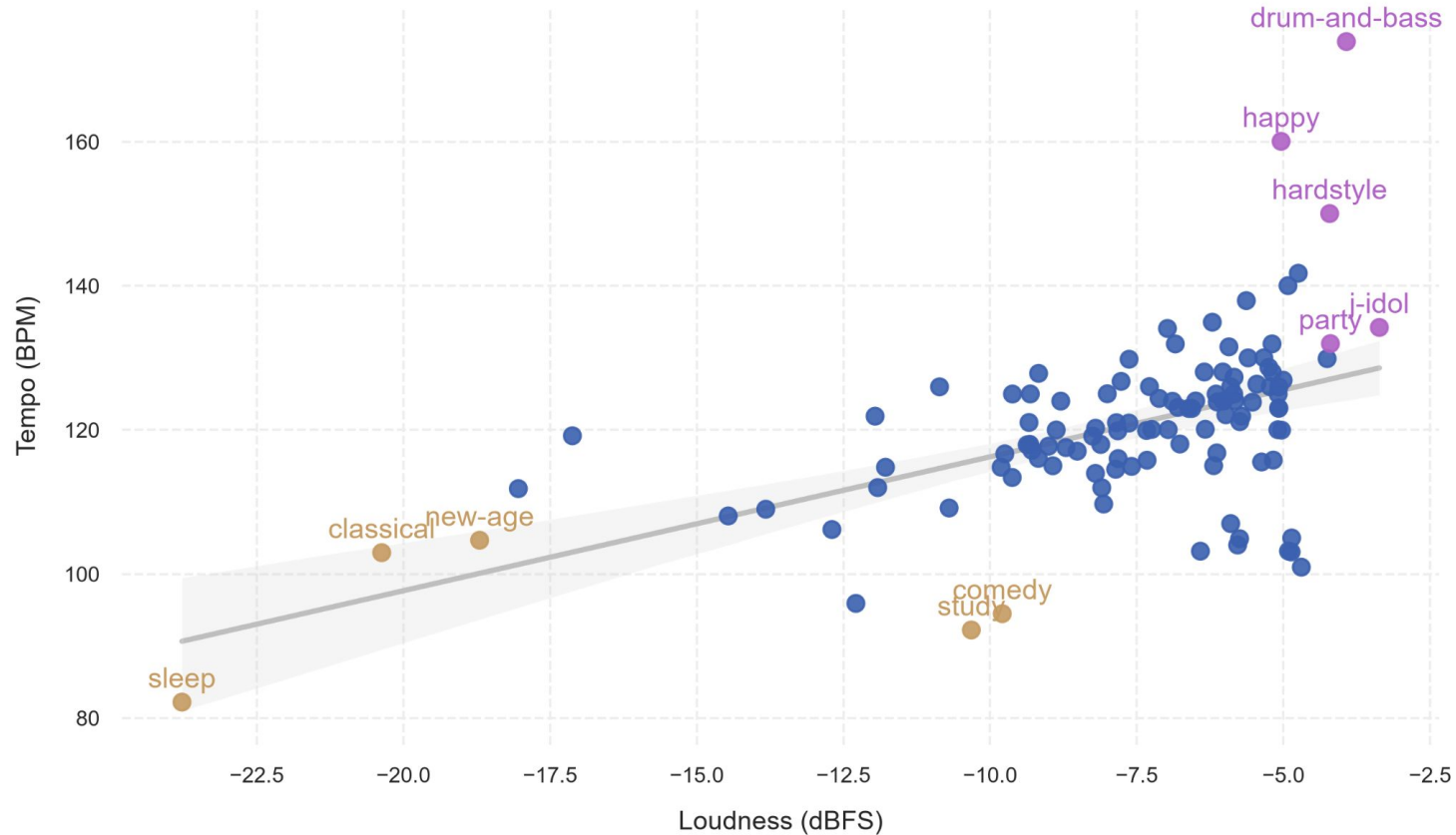
Loudness



Distribution of Track Loudness on Spotify



Louder Genres, Faster Tempos



Univariate Analysis – Artists (categorical)

31,347

unique artists in
the dataset

Goals:



Catalog diversity

How spread out is artist representation across the catalog?



Frequency distribution

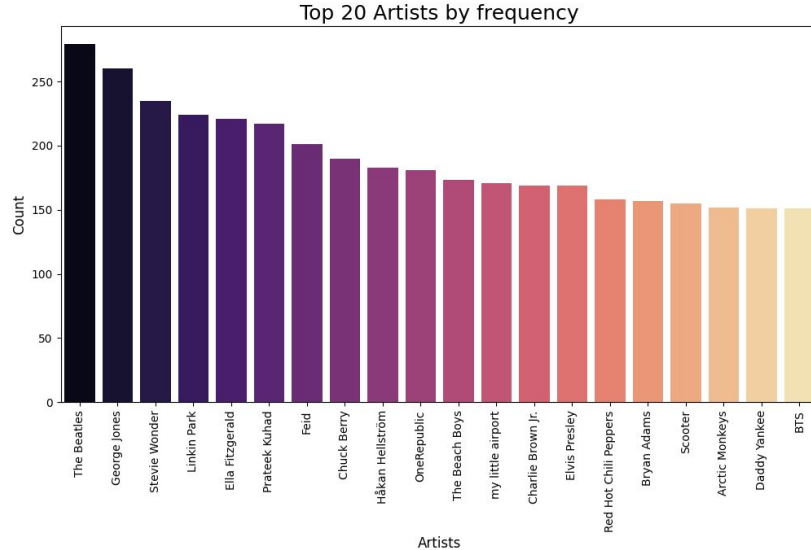
Which artists appear most often in the dataset?



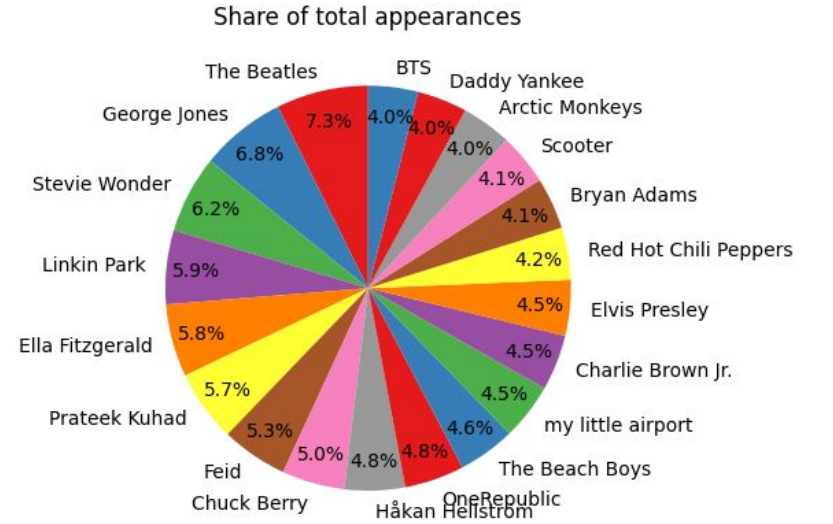
Proportional contribution

How much does each top artist account for overall?

Artists- Distribution

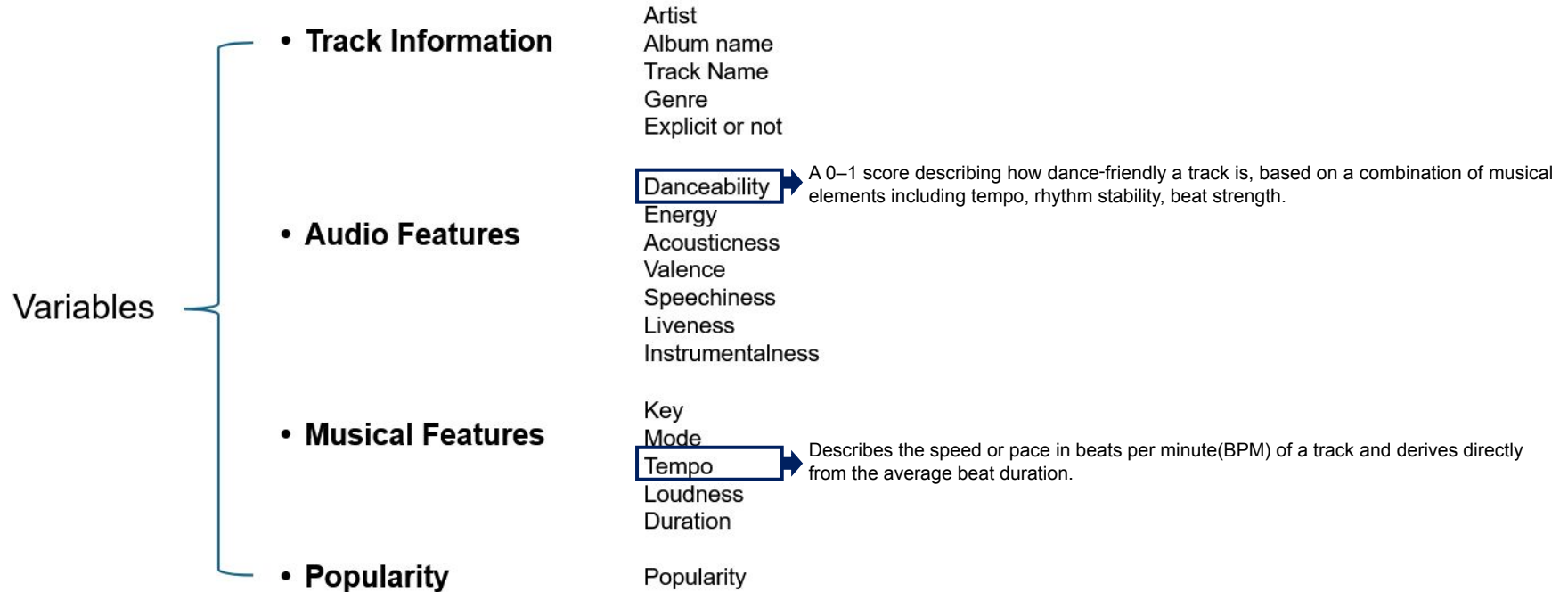


- The Beatles top the dataset with 280+ appearances — nearly double the artist ranked 20th



- Even in the top 20 artists list, each artist hold under 8% of appearances, pointing to a long-tailed, highly diverse catalog.

Bivariate EDA - continuous vs continuous



Bivariate EDA – tempo and danceability

Does a track's tempo influence how danceable it is?



Why these two variables

Both are continuous audio features; tempo is often assumed to drive perceived danceability.



Approach

Check distributions, clean invalid and outlier values, then examine the relationship



Questions to answer

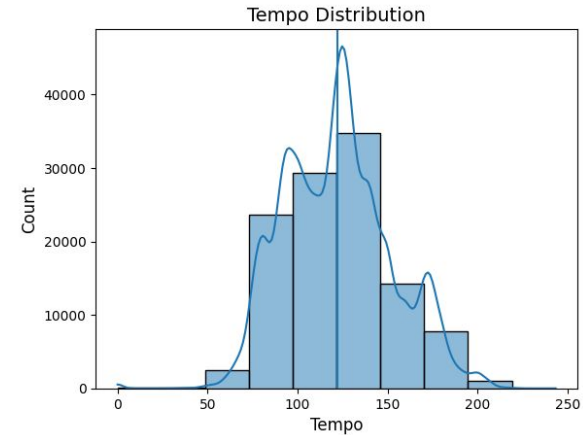
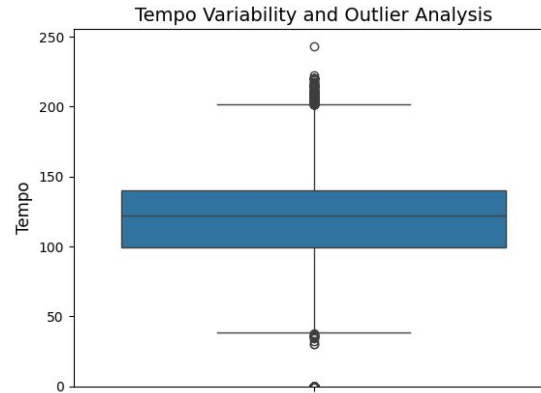
Is there a meaningful correlation, or are the two features largely independent?

Data quality & cleaning - tempo

```
num.tempo.describe()
```

	tempo
count	113550.000000
mean	122.175888
std	29.972861
min	0.000000
25%	99.296500
50%	122.020000
75%	140.073750
max	243.372000

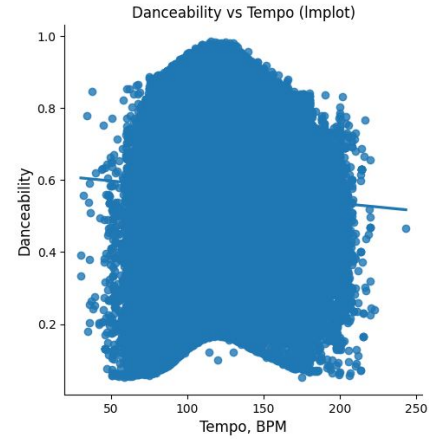
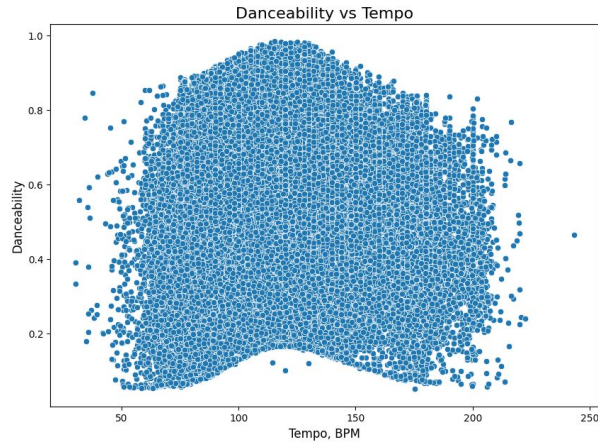
dtype: float64



Observations:

- Mean and median are approximately equal.
- Minimum tempo is **0 BPM**, which is not a musically valid value; most likely a data quality error.
- Boxplot confirms values below **~50 BPM** and above **~200 BPM** sit outside the normal range, including the 0 BPM error.
- Distribution is centered around **122 BPM** (mean).

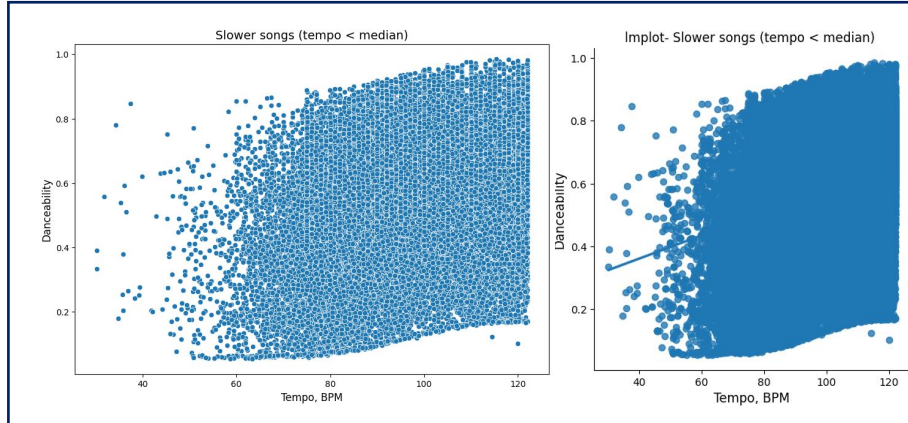
Correlation - tempo and danceability



Observations:

- Pearson correlation and spearman correlation is **-0.07** indicating that there is weak correlation between danceability and tempo.
- The scatterplot confirms this: danceability is spread broadly across the full tempo range, with no visible linear or monotonic pattern.

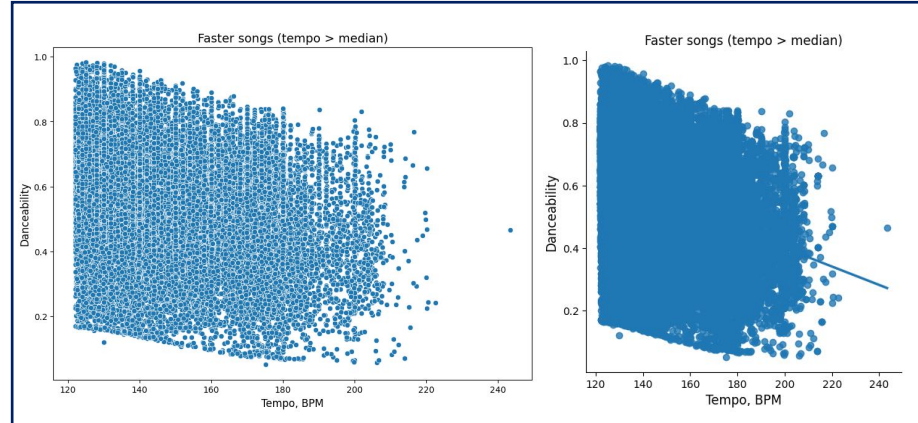
Correlation- Splitting tempo



For songs with tempo below the median (122 BPM):

Pearson & Spearman Correlation = 0.3

Danceability tends to **increase** as tempo rises.



For songs with tempo above the median (122 BPM):

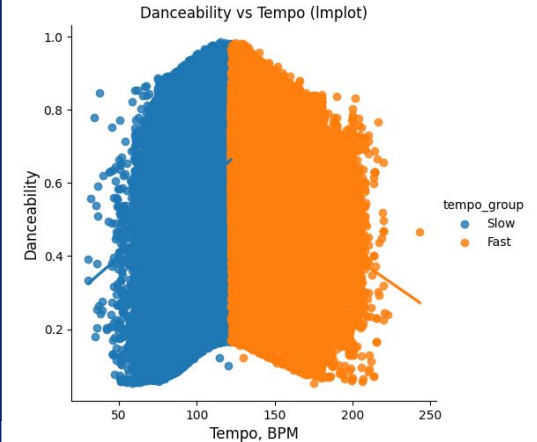
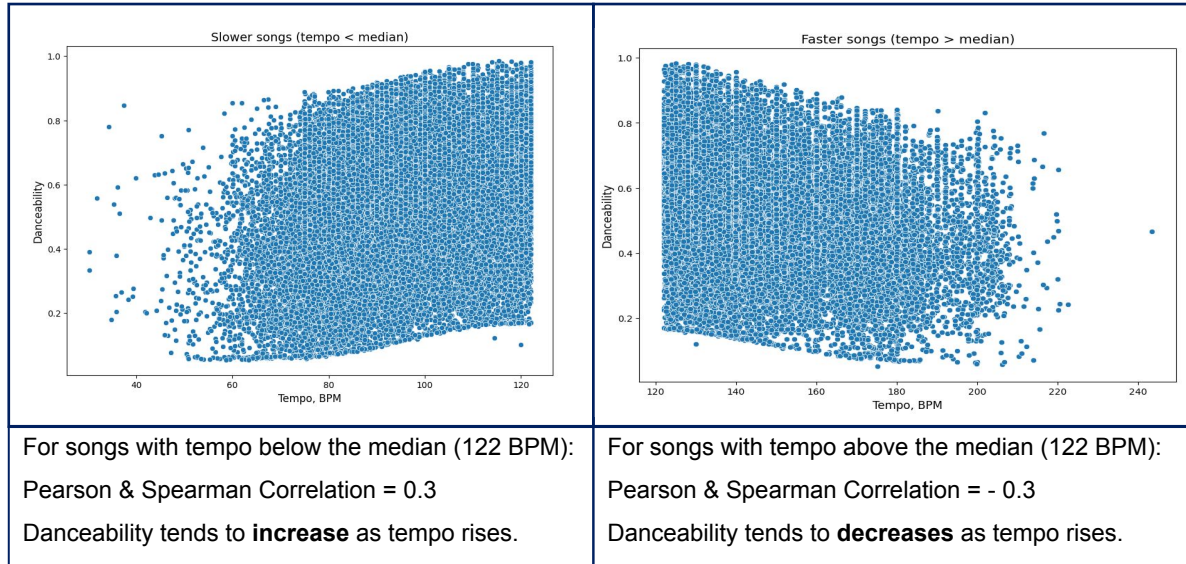
Pearson & Spearman Correlation = - 0.3

Danceability tends to **decreases** as tempo rises.

A case of Simpson's paradox:

- occurs when trends appears weak or absent in the overall dataset, but a clear emerges once the data is split into meaningful subgroups.
- Splitting at the median revealed two opposite trends - which cancel each other out when combined.
- The aggregate correlation alone would have wrongly suggested tempo has no effect on danceability.

Correlation- Splitting tempo



A case of Simpson's paradox:

- occurs when trends appears weak or absent in the overall dataset, but a clear emerges once the data is split into meaningful subgroups.
- Splitting at the median revealed two opposite trends - which cancel each other out when combined.
- The aggregate correlation alone would have wrongly suggested tempo has no effect on danceability.

Variables

- **Track Information**

Artist
Album name
Track Name
Genre
Explicit or not

- **Audio Features**

Danceability
Energy
Acousticness
Valence
Speechiness
Liveness
Instrumentalness

A measure from 0.0 to 1.0 describing the musical positiveness conveyed by a track. Tracks with high valence sound more **positive** (e.g. happy, cheerful, euphoric), while tracks with low valence sound more **negative** (e.g. sad, depressed, angry)

- **Musical Features**

Key
Mode
Tempo
Loudness
Duration

Business Questions:

- What is the impact of the word “love” in track names on valence.
- Does it have a positive connotation?

- **Popularity**

Popularity

Univariate Analysis - Valence

Numerical Variable:

- Measures of centrality: Mean, median
- Measures of dispersion: Variance, standard deviation, minimum, maximum, range, quantiles
- Shape of the distribution: Symmetry and kurtosis
- Visualizations: Histograms, box plots

```
count    113550.000
mean      0.474
std       0.259
min       0.000
25%       0.260
50%       0.464
75%       0.683
max       0.995
Name: valence, dtype: float64
```

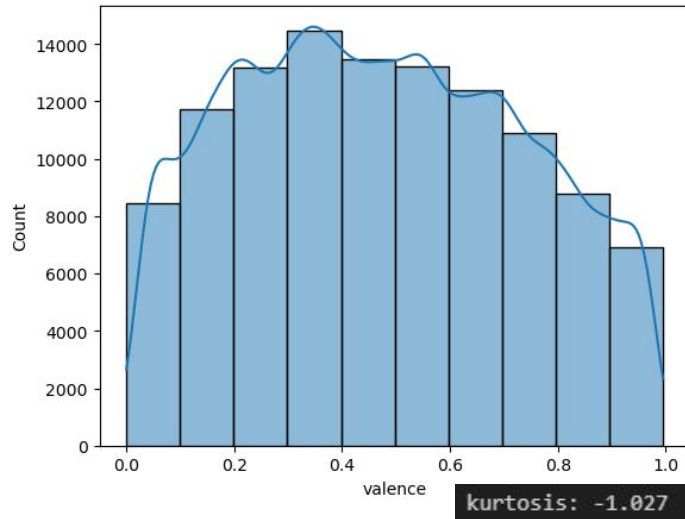
```
std: 0.259
var: 0.067
skewness: 0.115
min      0.000
25%      0.260
50%      0.464
75%      0.683
max      0.995
```

- $\text{std}/\text{mean} > 50\%$: values can vary significantly from the average
- max: 0.995 and min: 0. range of this variable (0-1)
- $\text{IQR} = Q3 - Q1 = 0.423$. 50% of the values are between 0.260 and 0.683
- skewness > 0 , distribution is positively skewed (right-skewed). long tail on the right side

median is slightly lower than the mean, suggesting skew in the distribution of Valence towards higher values (positiveness).

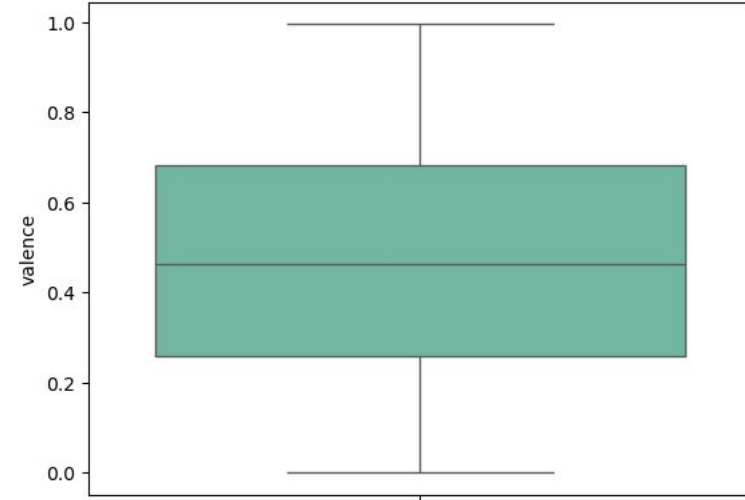
Shape of the distribution and Visualizations

Histogram



- tail on the right side, confirming our earlier inference from the skewness value
- negative kurtosis, the tails are not heavy and there is significant dispersion, the values are not concentrated near the mean.

Boxplot

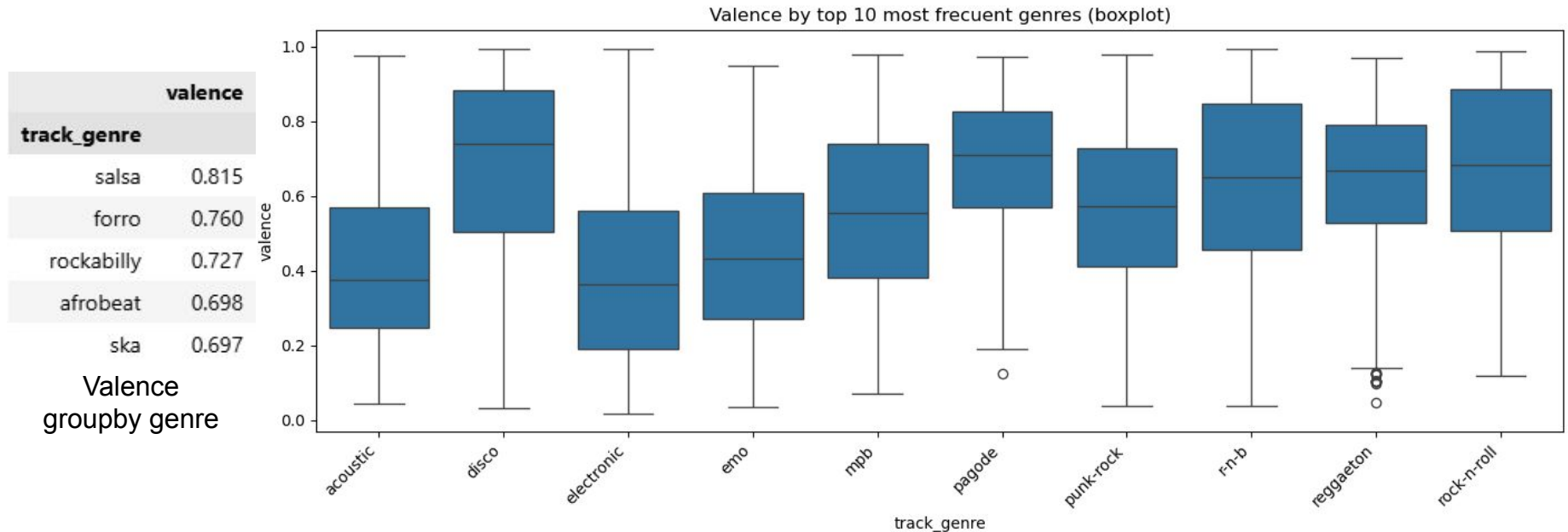


- values between 0 and 1
- no outliers for the univariate analysis of valence
- $IQR = Q3 - Q1 = 0.423$. 50% of the values are between 0.260 and 0.683

Bivariate Analysis - Valence

Numerical vs Categorical variable: Valence vs “love” in track_name

- Valence vs Genre: Bloxplot, groupby



Bivariate Analysis - Valence

Numerical vs Categorical variable: Valence vs “love” in track_name

- “love” in track_name vs Genre (Cat vs Cat): Crosstab

Excluding common articles and prepositions, the word “You” is the most popular word in song titles, appearing in over 5.8 million titles. It is followed closely by “Love” and the pronoun “I”.  Word Counter +1

index		track_name	loveintrack	valence
0	0	Comedy	False	0.715
1	1	Ghost - Acoustic	False	0.267
2	2	To Begin Again	False	0.120
3	3	Can't Help Falling In Love	True	0.143
4	4	Hold On	False	0.167

- Association with any genre?
- “love” = positive ?
- H0 : Valence has no association with word “love” in track names
- H1 : Valence has positive association with word “love” in track names

Bivariate Analysis - Valence

- “love” in track_name vs Genre (Cat vs Cat): Crosstab

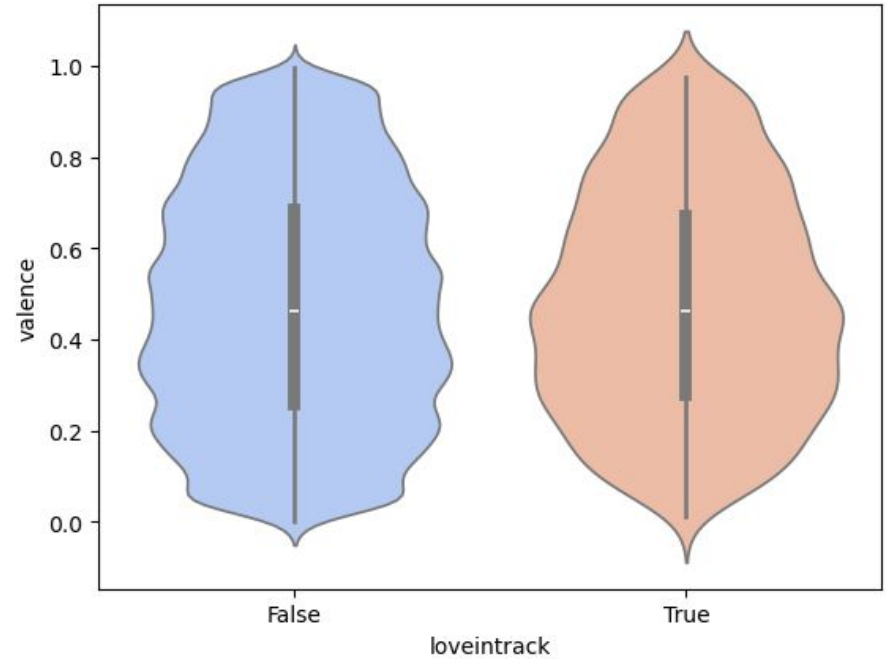
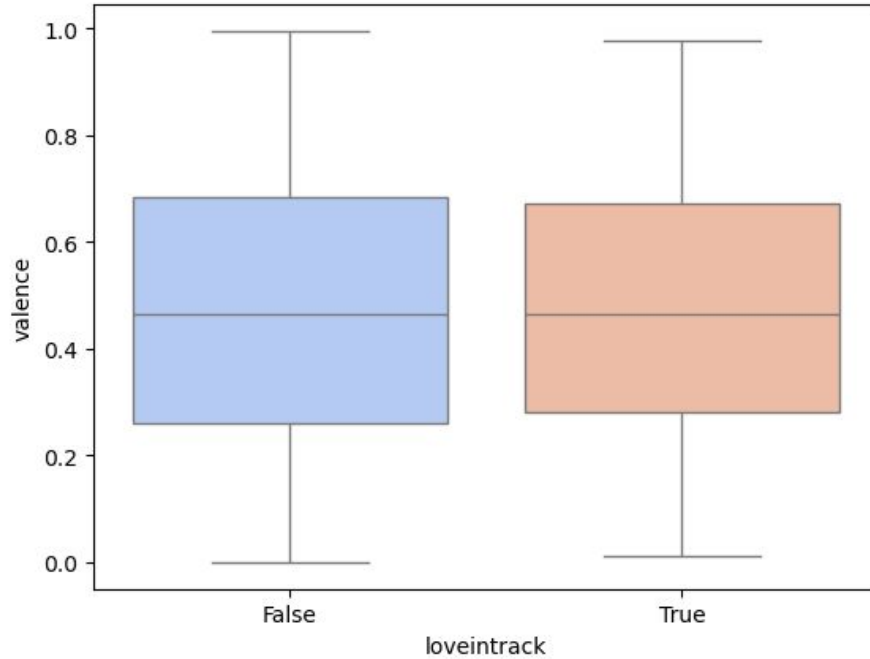
“love” groupby genre

	loveintrack
track_genre	
edm	90
chicago-house	74
synth-pop	69
dance	63
house	62

track_genre	acoustic	afrobeat	alt-rock	alternative	ambient	anime	black-metal	bluegrass
loveintrack								
True	61	12	10	15	16	10	4	21
False	939	987	989	984	983	989	993	977
reggaeton	rock	rock-n-roll	rockabilly	romance	sad	salsa	samba	sertanejo
0	14	40	57	4	45	0	2	0
1000	986	960	942	900	955	998	998	1000
reggaeton	rock	rock-n-roll	rockabilly	romance	sad	salsa	samba	sertanejo
0	14	40	57	4	45	0	2	0
1000	986	960	942	900	955	998	998	1000

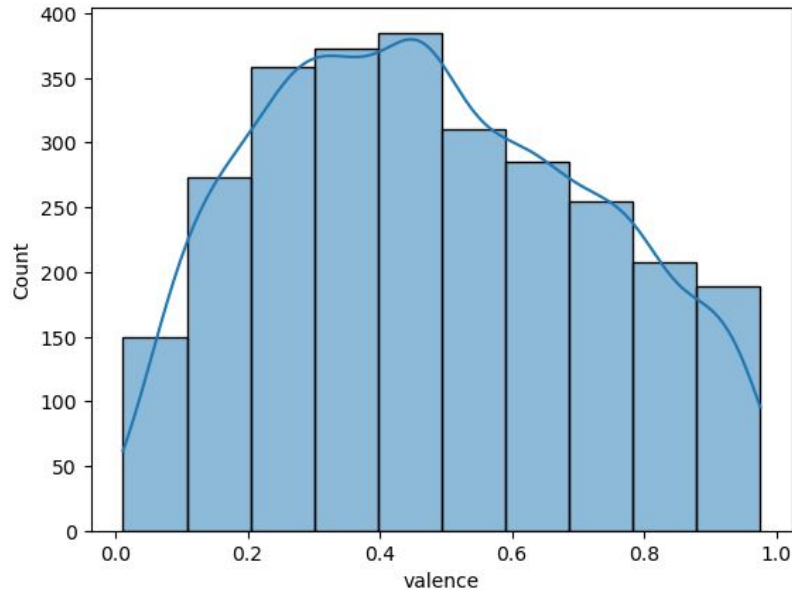
Bivariate Analysis - Valence

- Valence vs “love” in track_name: Bloxplot, violinplot



Bivariate Analysis - Valence

- Valence vs “love” in track_name: Histogram



- H0 : Valence has no association with “love” in track names
- H1 : Valence has positive association with “love” in track names
- H2 : Valence has negative association with “love” in track names

Univariate Analysis

Variables

- **Track Information**

Artist
Album name
Track Name
Genre
Explicit or not

- **Audio Features**

Danceability
Energy
Acousticness
Valence
Speechiness
Liveness
Instrumentalness

- **Musical Features**

Key
Mode
Tempo
Loudness

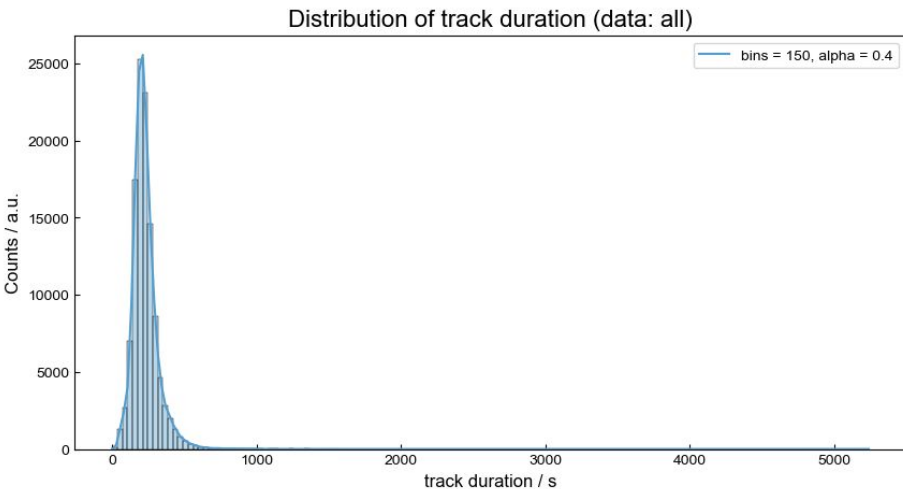
Duration

in ms, should be converted to seconds

- **Popularity**

Popularity

Univariate analysis: track duration



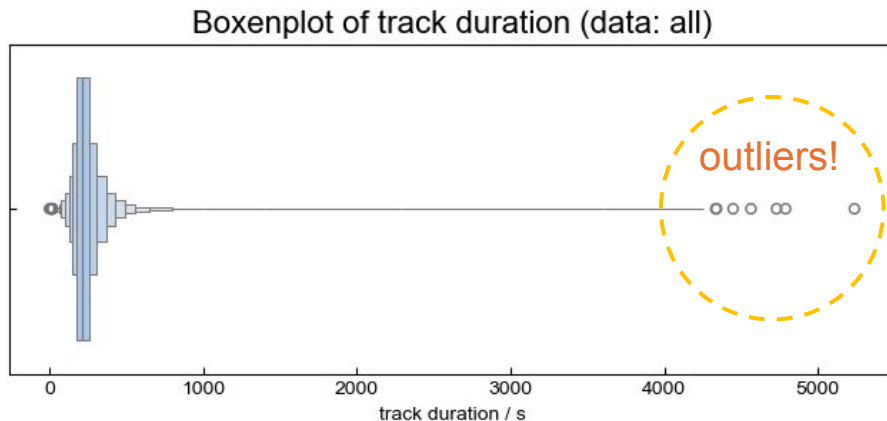
Remove outliers:
Calculate **IQR** (Interquartile **R**ange)

$$Q_1 - 1.5 * IQR < x < Q_3 + 1.5 * IQR$$

lower bound upper bound

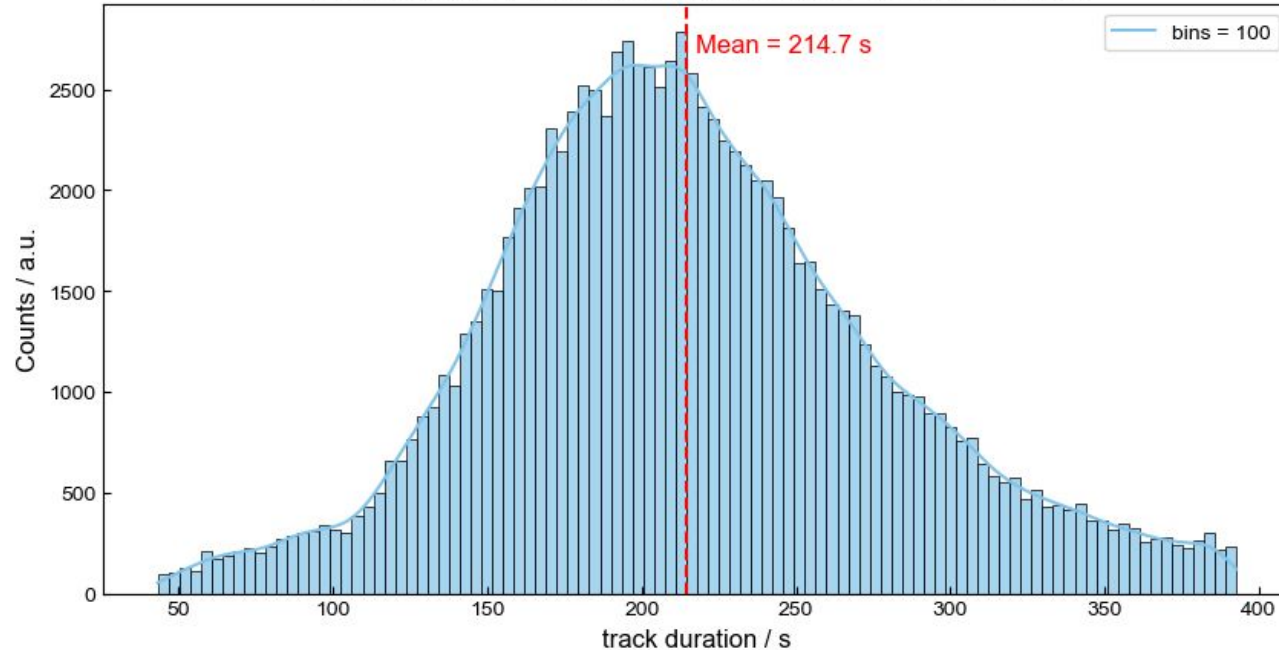
	index	duration / s
0	count	113549.0
1	mean	228.0
2	std	106.0
3	min	9.0
4	25%	174.0
5	50%	213.0
6	75%	262.0
7	max	5237.0

- How long is a song usually?
- What does distribution express?
→ definitely extreme long songs at end



Univariate analysis: track duration

Distribution of track duration (data: outlier removed)

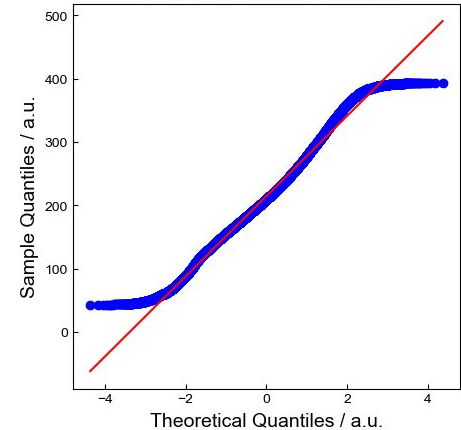


After removed outliers:

- Close to a normal distribution, slightly right skewed

	index	duration / s
0	count	107957.0
1	mean	215.0
2	std	64.0
3	min	43.0
4	25%	172.0
5	50%	210.0
6	75%	253.0
7	max	393.0

Q-Q Plot of Track Duration



Bivariate Analysis

Variables

- **Track Information**

Artist
Album name
Track Name
Genre
Explicit or not

- **Audio Features**

Danceability
Energy
Acousticness
Valence
Speechiness
Liveness
Instrumentalness

- **Musical Features**

Key
Mode
Tempo
Loudness
Duration

- **Popularity**

Popularity

Bivariate analysis: explicit vs. genre

→ What is explicit?



champagne problems 
evermore – Taylor Swift

Indicates whether a track contains explicit lyrics.

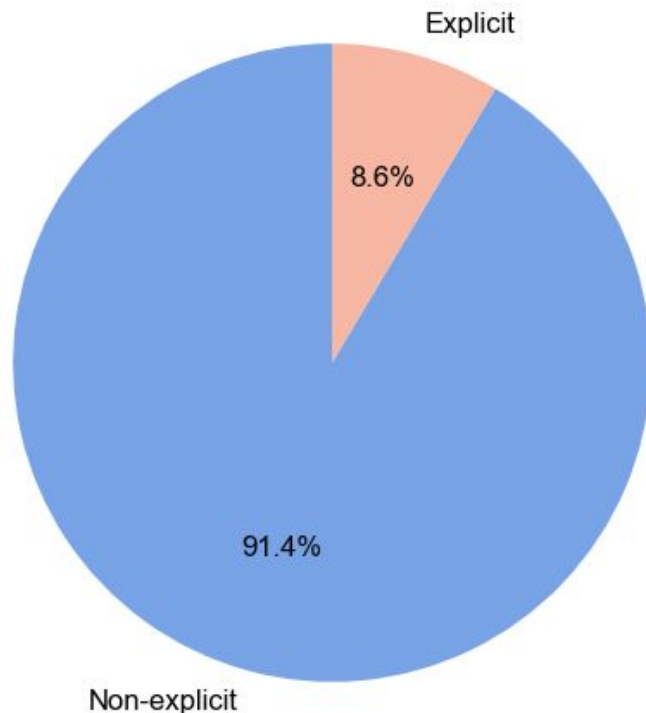
- **True (1)** : Explicit language or mature themes.
- **False (0)**: No explicit content.

→ General?

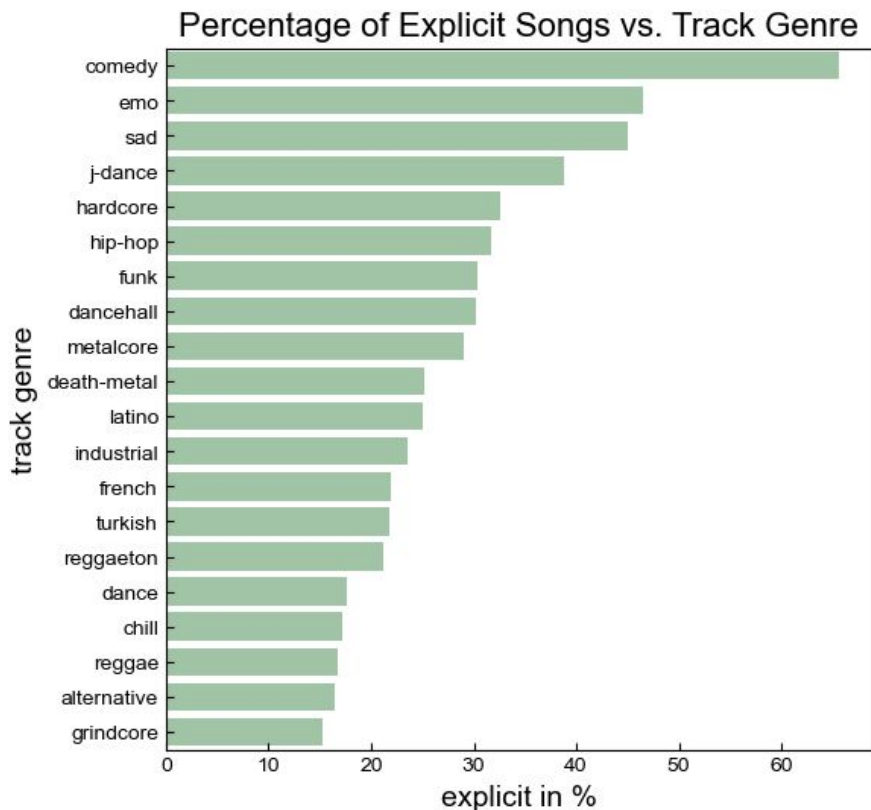
- **8.6 %** among this dataset

→ Bivariate analysis: Explicit vs. genre

Percentage of Explicit Songs



Bivariate analysis: explicit vs. genre



$$\text{explicit in \%} = \frac{\text{Count (True)}}{\text{Count (True)} + \text{Count (False)}} * 100$$

- **Comedy** even more than 50 % of the songs are with explicit content in lyrics.
- E.g. **hip-hop**, **metalcore** etc. are not surprising.

Bivariate Analysis

Variables

- **Track Information**

Artist
Album name
Track Name
Genre
Explicit or not

- **Audio Features**

Danceability
Energy
Acousticness
Valence
Speechiness
Liveness
Instrumentalness

- **Musical Features**

Key
Mode
Tempo
Loudness
Duration

- **Popularity**

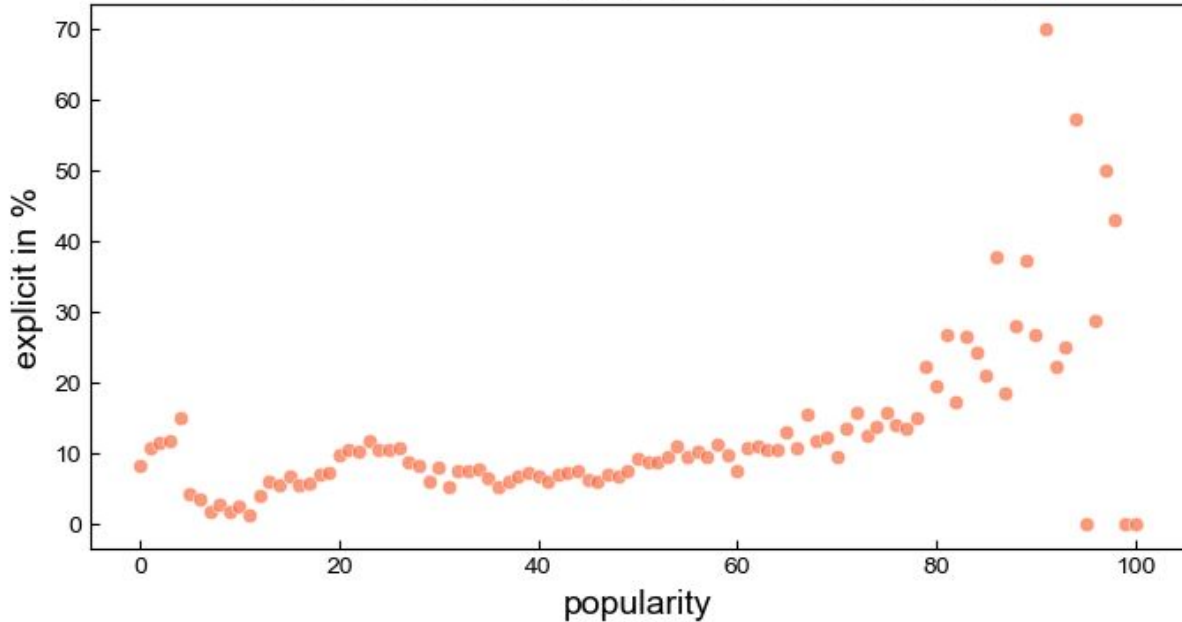
Popularity

Popularity: value between [0, 100]

Higher : more popular

Bivariate analysis: explicit vs. popularity

Percentage of Explicit Songs vs. Popularity



→ Popularity  explicit % **slightly**

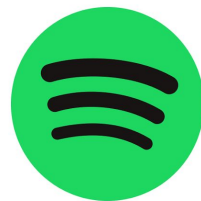

→ Among the most popular songs:
Value highly unstable

→ Hard to determine the relationship

Popularity: [0, 100]

$$\text{explicit in \%} = \frac{\text{Count (True)}}{\text{Count (True)} + \text{Count (False)}} * 100$$

Challenges & Outlook



- Importance of data **selection** and **cleaning**
- How to convert the lesson to practical work
- Choose the proper type of plot to present better
- Determine the outlier and remove
- Without precise correlation is also a result :)
- Collaboration!! Important!!!

THANKS!

Backup



Univariate Analysis



Bivariate Analysis